Zhongzhi Li

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Education

University of Georgia — Georgia, US
Ph.D. in Computer Science — *Aug 2025 – May 2030 (expected)*
Core courses: Efficient Deep Learning, Algorithms, and Internet of Things Security

Fudan University — Shanghai, China
M.Eng. in Aerospace Science and Technology; ***GPA: 3.41/4.00*** — *Sep 2022 – Jun 2025*
Core courses: Self-repairing flight control system, Atmospheric flight dynamics, and Aircraft system identification

Shenyang Aerospace University — Shenyang, China
B.E. in Computer Science and Technology; ***GPA: 8.28/10 (92/100)*** — *Sep 2018 – Jun 2022*
Core courses: Principles of Computer Organization, Operating Systems, and Data structures and algorithms

Research projects & Interests

My research makes AI systems reliable when data are scarce and mistakes are costly. I started with fault diagnosis for safety-critical systems, including aircraft sensors, power distribution networks, bearings, and insulated overhead conductors, where labeled failures are rare and every alarm must be explainable; four of these papers have each been cited 50+ times. Building those detectors taught me two lessons that shape my current work: a model is only as good as the coverage of its training data, and reliability has to be measured rather than assumed. The first led to data synthesis for LLM post-training (Less is Enough, ICML 2026 Oral) and for long-horizon agents (Recursive Synthesis); the second led to Long-Horizon-Terminal-Bench, which measures how far AI agents get on long, multi-step tasks.

1. **Less is Enough: Synthesizing Diverse Data in LLM Feature Space with Sparse Autoencoders** (co-first author, lis
   - Data diversity is crucial for effective LLM post-training, but common metrics (e.g. Distinct-n, Cosine Distance) often fail to reflect downstream task performance. We proposed Feature Activation Coverage (FAC) as a superior diversity metric that better predicts task performance by measuring coverage of task-relevant features.
   - Guided by FAC, we synthesize a small amount of targeted data to fill coverage gaps, achieving superior performance on Toxicity Detection, Reward Modeling, Behavior Steering, and Instruction Following. Notably, on Instruction Following, our method matches prior SOTA MAGPIE while using 150× less data. Project page: `Less is Enough`.
2. **Long-Horizon-Terminal-Bench (LHTB)** (core contributor, i.e., co-first author)
   - Binary pass/fail metrics cannot show how far an AI agent gets on long, multi-step work. LHTB evaluates agents on 46 long-horizon terminal tasks across 9 categories and scores partial progress with dense rewards.
   - Used to evaluate frontier models from xAI (Grok), Anthropic (Claude), OpenAI (GPT), and Google (Gemini); when Grok 4.5 reached the top, **Elon Musk** posted the result and named LHTB (about 13M combined views). #1 Hugging Face Daily Paper; 68K+ dataset downloads; listed in UK AISI's Inspect evaluation catalog. arXiv:2607.08964.

Intern Experience

Tencent America LLC (Hunyuan LLM Frontier) — Bellevue, WA, US
Research Intern (Agent) — *May 2026 – Aug 2026*

- **Problem:** agent benchmarks use short tasks and pass/fail scoring, so they cannot separate frontier models on long, multi-step work; and training data for long-horizon agents are scarce and costly to produce.
- **Evaluation:** as a core contributor (co-first author) of Long-Horizon-Terminal-Bench (LHTB), designed and implemented 30 of its 46 long-horizon terminal tasks (and 12 of the 31 added in LHTB 2.0), graded with dense rewards so that partial progress counts.
- **Training data and RL:** as co-first author (listed first) of Recursive Synthesis, built a pipeline that synthesized 37,484 verified long-horizon terminal tasks and 327,189 agent trajectories at roughly $0.05 per task, and used the synthesized tasks and environments for reinforcement learning (RL) training of terminal agents. Released data downloaded 100K+ times, incl. copies republished by Prime Intellect, Ai2's TMax team, and Open Athena.
- **Follow-up work:** extended this line in two 2026 preprints, as co-first author and as a core contributor respectively, scaling terminal environments to scientific domains through self-supervision and scaling agent trajectories for complex tasks through recursive self-rewrite.
- **Result:** LHTB's leaderboard ranks 20+ frontier models, including Grok, Claude, GPT, and Gemini; when Grok 4.5 topped it, Elon Musk posted the result and named LHTB (about 13M combined views). LHTB and Recursive Synthesis each ranked #1 on Hugging Face Daily Papers.

X Square Robot Shenzhen, China

Robotics Algorithm Intern *May 2025 – Jul 2025*

- **Problem:** diffusion-based Vision-Language-Action (VLA) policies generate robot trajectories through many denoising steps, which is too slow for real-time robot control.
- Benchmarked four policy acceleration methods and applied model distillation to cut the VLA policy's inference cost.

- Built a TensorRT deployment pipeline (PyTorch to ONNX to TensorRT, with per-module unit tests) and verified numerical parity with PyTorch by fixing sampling randomness.
- **Result:** sped up policy inference by about 2–3×, and delivered a ready-to-use TensorRT image with conversion scripts and documentation, integrated into the team's testing and CI/CD workflow.

Selected Publication & Academic Services

47 publications (22 as first or co-first author). Google Scholar: 723 citations, h-index 14, i10-index 17 (Nov 2026), plus 69 citations of Chinese-language papers in CNKI; four papers each cited 50+ times.

[1] **Li Z***, Wu X*, et al. Less is enough: synthesizing diverse data in feature space of LLMs. *ICML*, 2026. **(Oral, top 0.7%)**

[2] **Li Z**, Han J, et al. Beyond Prompts: Discovering Jailbreak Motifs in Sparse Feature Space. *EMNLP* (Main), 2026.

[3] Li Z*, **Li Z***, Shi Y*, et al. Long-Horizon-Terminal-Bench: Testing the Limits of Agents on Long-Horizon Terminal Tasks with Dense Reward-Based Grading. *arXiv:2607.08964*, 2026.

[4] **Li Z***, Shi Y*, Li Z*, et al. Recursive Synthesis for Long-Horizon Terminal Tasks. *arXiv:2608.05466*, 2026.

[5] **Li Z***, Shi Y*, Li Z*, et al. Self-Supervised Scaling of Terminal Environments for Scientific Domains. *arXiv:2610.02710*, 2026.

[6] **Li Z**, Tu J, Zhu J, et al. Scalable and Reliable Deep Transfer Learning for Intelligent Fault Detection via Multi-Scale Neural Processes Embedded with Prior Knowledge. *Neurocomputing*, 2025: 131364.

[7] **Li Z**, Zhang Y, Ai J, et al. A Lightweight and Explainable Data-Driven Scheme for Fault Detection of Aerospace Sensors. *IEEE Transactions on Aerospace and Electronic Systems*, 2023, 59(6): 8392-8410.

[8] Qu N, **Li Z**, Li X, et al. Multi-Parameter Fire Detection Method Based on Feature Depth Extraction and Stacking Ensemble Learning Model. *Fire Safety Journal*, 2022, 128: 103541.

[9] **Li Z**, Huang Y, Li X, et al. Wildland Fire Burned Areas Prediction Using Long Short-Term Memory Neural Network with Attention Mechanism. *Fire Technology*, 2021, 57: 1-23.

* *Co-first authors.*

Academic service: Contributed extensive peer-review service, completing 48 rounds of manuscript review for venues including *ICLR 2027*, *WSDM 2027*, *IEEE Transactions on Artificial Intelligence*, *IEEE Transactions on Industrial Informatics*, and *IEEE Internet of Things Journal*.

Awards & Achievements

Postgraduate period Shanghai, China

*Received **6** scholarships* *Sep 2022 – Jun 2025*

- Qingyun Scholarship (*Awarded annually to only 10 graduate students*) Oct 2024
- National Scholarship *(Only 1 scholarship recipient is selected from 45 students)* Dec 2023
- Yang Yongming Scholarship Dec 2023
- Inspirational Scholarship of Zhang Mingwei Dec 2022

Undergraduate period Shenyang, China

*Received **12** scholarships and **10** national level competition awards* *Sep 2018 – Jun 2022*

- National Scholarship (*Only 1 recipient is selected from 155 students*) Dec 2020
- Hexagon Enterprise Scholarship Dec 2021
- Academic Scholarship of First Class Jun 2021
- Academic Scholarship of First Class Dec 2021
- School Annual Personality Scholarship (*Awarded annually to only 10 undergraduate students*) Dec 2021
- National Physics Academic Competition **Second Prize** Aug 2019
- National Student Mathematics Competition **Second Prize** Nov 2019
- National Students Asia Pacific Mathematical Modeling Competition **Second Prize** Dec 2019
- National Students' May Day Mathematical Modeling Competition **First Prize** Jun 2020
- National Student Mathorcup Mathematical Modeling Competition **First Prize** ($\approx 5\%$) Jun 2020